

HIGH PERFORMANCE COMPUTING

Tugas 2



Disusun oleh:

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FAKULTAS MATEMATIKA DAN ILMU PENGETAHUAN ALAM

Program Studi S-1 Teknik Informatika

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Source Code: <https://github.com/WilsonnnTan/hpc-assignment/tree/master/assignment-2>

Spin up cluster:

```
PS C:\College_Computer_Science\Sem 7\high-performance-computing\assignment-2> docker-compose up --build
[+] Running 2/2
X computenode2 Error pull access denied for cluster-node, repository does not exist or may require 'docker login...' 1.4s
X computenode1 Error pull access denied for cluster-node, repository does not exist or may require 'docker login...' 1.4s
[+] Building 46.9s (13/13) FINISHED docker:desktop-linux
=> [headnode internal] load build definition from Dockerfile 0.0s
=> => transferring dockerfile: 1.48kB 0.0s
=> [headnode internal] load metadata for docker.io/library/ubuntu:22.04 0.4s
=> [headnode internal] load .dockerignore 0.0s
=> => transferring context: 2B 0.0s
=> CACHED [headnode 1/7] FROM docker.io/library/ubuntu:22.04@sha256:b8b6ee6aa931ecd9d0d952abc34dc0e5f7c6a30c6bb71b079fe399fde0329c02 0.0s
=> => resolve docker.io/library/ubuntu:22.04@sha256:b8b6ee6aa931ecd9d0d952abc34dc0e5f7c6a30c6bb71b079fe399fde0329c02 0.0s
=> [headnode internal] load build context 0.0s
=> => transferring context: 35B 0.0s
=> [headnode 2/7] RUN apt-get update && apt-get install -y --no-install-recommends openssh-server openssh-client 39.8s
=> [headnode 3/7] RUN echo 'root:cluster123' | chpasswd 0.3s
=> [headnode 4/7] RUN sed -i 's/#\?PermitRootLogin.*/PermitRootLogin yes/' /etc/ssh/sshd_config && sed -i 's/#\?PasswordAuthenticat 0.3s
=> [headnode 5/7] RUN mkdir -p /root/.ssh && ssh-keygen -t rsa -b 4096 -f /root/.ssh/id_rsa -N "" && cat /root/.ssh/id_rsa.pub 1.2s
=> [headnode 6/7] COPY entrypoint.sh /entrypoint.sh 0.1s
=> [headnode 7/7] RUN chmod +x /entrypoint.sh 0.3s
=> [headnode] exporting to image 4.0s
=> => exporting layers 3.0s
=> => exporting manifest sha256:f0703cc6e7f0e86e18d2c192f37742a2597c7bb675f18077e388df972858f5bb 0.0s
=> => exporting config sha256:cf2568b7a072f74c85af05630d8d6fd4e5aa615a0d0639a61081d6dff3cee3d2 0.0s
=> => exporting attestation manifest sha256:7815b5130c37191142a39a8bd87060af5336a1d701172b55cfb82e903c8a4688 0.1s
=> => exporting manifest list sha256:1053b33306520760a827eae50bca7fbb4bba72112b897bf82695e388b362a4b3 0.0s
=> => naming to docker.io/library/cluster-node:latest 0.0s
=> => unpacking to docker.io/library/cluster-node:latest 0.8s
=> [headnode] resolving provenance for metadata file 0.0s
[+] Running 5/5
✓ Network assignment-2_cluster-net Created 0.1s
✓ Volume "assignment-2_shared-data" Created 0.0s
✓ Container headnode Created 2.6s
✓ Container computenode2 Created 0.2s
```

SSH to headnode from powershell:

```
PS C:\College_Computer_Science\Sem 7\high-performance-computing> ssh root@localhost -p 2222
root@localhost's password:
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.18.33.2-microsoft-standard-WSL2 x86_64)
```

```
* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:        https://ubuntu.com/pro
```

This system has been minimized by removing packages and content that are not required on a system that users do not log into.

To restore this content, you can run the 'unminimize' command.

The programs included with the Ubuntu system are free software; the exact distribution terms for each program are described in the individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by applicable law.

SSH to computenode1 from headnode:

```
root@headnode:~# ssh root@172.28.0.11
Warning: Permanently added '172.28.0.11' (ED25519) to the list of known hosts.
root@172.28.0.11's password:
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.18.33.2-microsoft-standard-WSL2 x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

This system has been minimized by removing packages and content that are
not required on a system that users do not log into.

To restore this content, you can run the 'unminimize' command.

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individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

root@computenode1:~# exit
logout
Connection to 172.28.0.11 closed.
```

SSH to computenode2 from headnode:

```
root@headnode:~# ssh root@172.28.0.12
Warning: Permanently added '172.28.0.12' (ED25519) to the list of known hosts.
root@172.28.0.12's password:
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.18.33.2-microsoft-standard-WSL2 x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

This system has been minimized by removing packages and content that are
not required on a system that users do not log into.

To restore this content, you can run the 'unminimize' command.

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

root@computenode2:~# exit
logout
Connection to 172.28.0.12 closed.
```